

Marked circuits and fixed-five rooted interfaces in cubic graphs

Structural reductions and finite censuses

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Abstract

Let H be a finite simple cubic graph and let S be a four-edge matching. As algebraic setup, we first identify five Eulerian edge sets covering every edge twice with an everywhere-anisotropic flow in a four-dimensional quadratic space over $\mathbb{F}_2$, including the exact convention for loops. We then record eight narrowly scoped results. First, if H has a Tait colouring in which the four marks have one common colour, and

$$|\delta_H(X)| + |S \cap E(H[X])| \geq 4$$

whenever $H[X]$ contains a cycle, then S is contained in an even edge set whose every circuit component contains an even number of marks. The proof combines elementary shore lemmas with three published prescribed-cycle and decomposition results. It does not require universal separation after the common-colour Tait colouring has been obtained.

Second, suppose H is cyclically 4-edge-connected, the marks are separated in a fixed common-colour Tait colouring, and a differently coloured edge $f \notin S$ is forbidden. If no cycle of $H - f$ contains all four marks, then f belongs to an independent cyclic four-edge cut. Hence a cycle through all four marks and avoiding f always exists when H is cyclically 5-edge-connected.

Third, for the surviving exceptional rooted three-pole interface in a separate $D_5 = \binom{[5]}{2}$ flow model, we prove a necessary cycle-support obstruction. If a nonbridge root has only one possible label after the connector triangle is fixed, every cycle through that root has a label support which spans all five coordinates and is either non-bipartite or the star $K_{1,4}$. An exact gluing factorization shows how this condition arises when a fixed-five boundary signature equals one of the exceptional type patterns in the literature. Missing root base pairs also force explicit odd proper-core cuts. These are necessary restrictions, not eliminations.

Fourth, we prove that a nonbridge root has one of the three base pairs whenever capping its three connectors by one vertex gives a Tait-colourable cubic graph. One such Tait cap excludes the equality-only and disjointness-only exceptional root relations; two Tait caps exclude all three exceptional relations. The mixed equality-plus-intersection relation is not excluded one-sidedly.

Fifth, two exact solvers show that every nonempty nonbridge rooted three-pole signature through order 17 contains one of three base pairs: 15 645 623 roots in 654 676 canonical cores are checked. Sixth, a root-end factor base pair lifts through an arbitrary cubic three-sum path as a three-label fork. Two such forks force both an unequal-intersection and a disjoint cross-pair. Together with two exact endpoint classifiers through factor order 26 and complete cyclically four-connected cap classifiers through cap order 26, this first gives a finite lower bound 28 for a bridge-free connected simple terminal-distinct fixed-five exceptional pole. An explicit-witness classification of all 9 725 709 independent-edge deletions from the 12 517 retained cyclically four-connected order-28 caps then raises the bound to 30. The argument is human-checkable except for the explicitly finite classifications.

Seventh, on the 31 cyclically 4-edge-connected bridgeless non-Tait cubic graphs of order 22, all 14 322 independent-edge deletion poles have the full ten-type fixed-five boundary signature.

The larger all-cap order-22 computation also checks all 5 956 104 independent-edge deletion poles and finds no exceptional signature.

Eighth, an exhaustive canonical computation finds no *Tait-colourable* connected simple cubic graph on 22 vertices having a universally separated four-edge matching. Two separately written enumeration methods agree on 7 174 735 colourable graphs and 95 360 112 Tait colourings modulo global colour permutation. This manuscript neither proves nor disproves the Five-Cycle Double Cover Conjecture.

AI-use disclosure. OpenAI Codex agents, under the direction of Atharva Vaidya, generated the initial arguments, wrote the census programs, executed and audited the computations, and drafted this manuscript. Other Codex agents independently rechecked selected proofs and enumerations; those are AI-agent checks, not independent human verification or peer review. All mathematical arguments used here are displayed for line-by-line human checking, and the computational claims are separated from them. The human author should independently verify the paper and establish priority before submission.

1 Introduction and exact scope

A binary cycle of a graph is an edge set with even degree at every vertex. In a cubic graph its nonempty connected components are cycles. The first part of this paper asks when four marked edges lie in a binary cycle whose components each contain an even number of marks. This marked parity question arose in a larger investigation of the Five-Cycle Double Cover Conjecture, but the results below do not resolve that conjecture.

For clarity, the standard conjecture asks for at most five Eulerian edge subsets whose incidence multiplicities cover each edge exactly twice [8, 6]. The orientable variant imposes additional directional conditions and is a different problem. Neither version is proved here. In particular, our unrooted theorem has a marked-cut hypothesis, our rooted theorem leaves a cyclic four-cut interface, and the census is restricted to one order, simple graphs, and Tait-colourable inputs. All D_5 boundary signatures below use one fixed five-coordinate universe. They are not the unbounded-colour relation of Máčajová, Mazzuoccolo, and Tabarelli; our finite computations do not classify colourings needing more than five coordinates.

The contributions

Result	Content	Nature of evidence
Theorem 4.1	Four common-colour marks satisfying the marked cyclic-cut inequality admit an even-marked binary cycle.	Human proof using explicitly identified imported theorems.
Theorem 5.1	A failed forbidden-edge reduction is localized to an independent cyclic four-cut; cyclic connectivity five closes the rooted case.	Human proof; only the four-independent-edge cycle theorem is imported.
Theorem 6.4	A nonbridge singleton root in the normalized D_5 three-pole interface forces every root cycle to have a spanning translation blocker.	Human algebraic and graph-cut proof; a necessary obstruction only.
Theorem 6.10	Failure of three rooted base pairs forces an odd proper-core cut carrying one common coordinate.	Elementary human proof; a necessary certificate only.
Theorem 6.12	A Tait-colourable one-vertex cap forces one rooted base pair; the exceptional consequence distinguishes one-sided from two-sided closure.	Elementary human proof with an explicit counterexample to the overstrong one-sided inference.
Theorem 7.3	Every nonempty nonbridge rooted three-pole signature through order 17 contains a base pair.	Two exact solvers with matching per-shard transcript digests.
Theorem 6.17	A base pair at an endpoint factor lifts through the entire three-sum path as a fork; two forks force intersection and disjointness.	Human proof, with an exhaustive 30^2 set-system replay.
Theorems 7.4 and 7.9	Endpoint factors through order 26 contain a base pair; combined with explicit order-28 cap witnesses this gives the scoped lower bound 30.	Two exact rooted classifiers, two exact cap classifiers, and a human reduction.
Theorem 7.8	Every retained cyclically four-connected order-28 cap deletion realizes two boundary orbits which jointly exclude all six exceptional masks.	19 451 418 explicit D_5 labellings checked by independent Python and C++ replays.
Theorem 7.5	Every order-22 independent-edge deletion in the cyclically four-connected non-Tait cap corpus has the full fixed-five signature; the complete all-cap corpus has no exceptional row.	Two full boundary classifiers plus the complete two-classifier exceptional-signature census.
Theorem 8.2	No Tait-colourable connected simple cubic graph of order 22 has a universally separated four-edge matching.	Exhaustive computation with two algorithmically different enumerators.

The human arguments appear to be useful deductions from known cycle and boundary-

signature machinery. The rooted factorizations and finite frontiers appear new to the authors, but the priority search performed for this draft was targeted rather than exhaustive.

A scope correction discovered during audit

An earlier project note stated the computational result over all connected simple cubic graphs of order 22. The programs, correctly for the motivating application, search only graphs having at least one Tait colouring. This qualifier is logically essential. If a graph has no Tait colouring, the assertion “separated in every Tait colouring” is vacuously true under the literal universal quantifier. Moreover, every cubic graph on 22 vertices has a matching of size at least four: a maximal matching of size m has endpoints dominating at most $6m$ vertices, so $m \geq \lceil 22/6 \rceil = 4$. Theorem 8.2 therefore states exactly the nonvacuous result established by the programs.

2 Definitions and imported results

All graphs are finite. Unless stated otherwise, they are simple. For $X \subseteq V(G)$, let $\delta_G(X)$ be the set of edges with exactly one end in X , and let $G[X]$ be the induced subgraph. A *cycle* is a connected 2-regular subgraph. A *binary cycle* is an edge subset $Q \subseteq E(G)$ such that every vertex has even degree in Q .

A graph is *cyclically k -edge-connected* if it has no edge cut of size less than k with a cycle on both shores. A *Tait colouring* of a cubic graph is a proper edge-colouring with three colours a, b, c . Every two colour classes induce a spanning 2-regular subgraph, called a *bichromatic factor*; its components are even cycles.

An exact elliptic-flow reformulation

For later use, put

$$\Gamma = \left\{ x \in \mathbb{F}_2^5 : \sum_{i=0}^4 x_i = 0 \right\}, \quad q(x) = \sum_{0 \leq i < j \leq 4} x_i x_j. \quad (2.1)$$

If $x \in \Gamma$, then

$$q(x) = \binom{\text{wt}(x)}{2} \pmod{2}.$$

Thus $q(x) = 1$ exactly for the ten vectors of weight two, which we identify with

$$D_5 = \binom{\{0, 1, 2, 3, 4\}}{2}. \quad (2.2)$$

The polar form is

$$B(x, y) = q(x + y) + q(x) + q(y) = \sum_{i=0}^4 x_i y_i \quad (x, y \in \Gamma). \quad (2.3)$$

It is nondegenerate: a vector orthogonal to every $e_i + e_j$ has all coordinates equal, and the all-one vector is not in Γ . The five nonzero singular vectors are $\mathbf{1} + e_i$, $0 \leq i < 5$. They span Γ , with their sum as their only dependence. Hence every isometry permutes them faithfully. Coordinate permutations already realize all five permutations, so

$$O(\Gamma, q) \cong S_5. \quad (2.4)$$

Proposition 2.1 (Elliptic-flow equivalence). *Let G be a finite multigraph. Five Eulerian edge subsets $C_0, \dots, C_4$ cover each edge exactly twice if and only if there is a map $\phi : E(G) \rightarrow \Gamma$ such that*

$$q(\phi(e)) = 1 \quad (e \in E(G)), \quad \sum_{\substack{\text{edge-ends} \\ \text{incident with } v}} \phi(e) = 0 \quad (v \in V(G)). \quad (2.5)$$

A loop is counted by its two edge-ends in the second sum.

Proof. Given the five edge subsets, let coordinate i of $\phi(e)$ be one exactly when $e \in C_i$. Exact double coverage makes $\phi(e)$ a weight-two vector, hence an element of $q^{-1}(1)$. Coordinate i of the vertex equation is precisely the degree in C_i modulo two. Conversely, $q(\phi(e)) = 1$ forces $\phi(e)$ to have exactly two nonzero coordinates. Defining $C_i = \{e : \phi(e)_i = 1\}$ therefore gives exact double coverage, and the five coordinate equations give the five even-degree conditions. The two constructions are inverse. $\square$

Parallel edges are distinct edge variables. A loop contributes twice to each selected degree, so it cancels in the characteristic-two vertex equation; counting a loop only once in a SAT/XOR incidence row would be an incorrect encoding. Empty C_i are allowed, so this five-coordinate formulation means “at most five” in the standard sense [8, 6]. It imposes no orientation condition. The underlying two-subset encoding also appears as a *CDC-colouring* in Máčajová, Mazzuoccolo, and Tabarelli [3, Definition 3.1]. Proposition 2.1 is an exact elementary reformulation, not a resolution or a priority claim; priority for the quadratic packaging has not been established.

Definition 2.2 (Separation). Let S be a matching in a Tait-coloured cubic graph. The marks are *separated in that colouring* if every bichromatic factor cycle contains at most one member of S . If the graph is Tait-colourable, then S is *universally separated* if it is separated in every Tait colouring.

We use the following three published results. They are the only non-elementary imported ingredients in the proof of Theorem 4.1.

- (I1) Knappe and Pitz [2, Fact 5.4] prove $g(3) = 3$. In the form used here: if G is 3-edge-connected and $F \subseteq E(G)$ has at most three edges, then a circuit contains F if and only if F contains no odd edge cut of G .
- (I2) Aldred, Ellingham, Hemminger, and Holton [1, Theorem 3.3] prove that a free edge system of size at most four in a quasi 4-connected graph lies on a cycle. In particular, four independent edges lie on a cycle. They also note that cyclically 4-edge-connected cubic graphs are quasi 4-connected.
- (I3) Every 3-connected cubic graph is obtained by repeated vertex 3-sums of cyclically 4-edge-connected cubic graphs. This is Theorem 6.1 of Nedela, Seifrtova, and Skoviera [5]; see also Pastor’s component-tree theorem [7].

Here a circuit in (I1) is a connected closed trail. In a cubic graph its edge-induced degrees are positive and even, hence equal to two, so its edge set is a cycle.

Lemma 2.3 (Tait cut parity). *In a Tait-coloured cubic graph, the numbers of cut edges of colours a, b, c have the same parity. In particular, the three edges of a three-edge cut have three distinct colours, and a Tait-colourable cubic graph has no bridge.*

Proof. For a colour p , counting incidences of the p -coloured perfect matching with X gives

$$|X| \equiv |\delta_G(X) \cap E_p| \pmod{2}.$$

This holds for all three colours. If the cut has size three, all three counts are odd and sum to three, so each is one. If a single edge were a cut, its colour count would have parity different from the other two. $\square$

Lemma 2.4 (Precolouring separated marks). *Let H be Tait-colourable and let S be universally separated. Every map $S \rightarrow \{a, b, c\}$ extends to a Tait colouring. In particular, there is a Tait colouring in which all marks have one common colour.*

Proof. Start with any Tait colouring. If a mark e currently has colour p and is requested to have colour q , interchange p and q on the unique pq -factor cycle through e . Universal separation says that this cycle contains no other mark, so all previously fixed marks keep their colours. Process the marks one at a time. $\square$

3 Two elementary shore lemmas

The marked hypothesis used throughout the first theorem is

$$|\delta_H(X)| + |S \cap E(H[X])| \geq 4 \quad \text{whenever } H[X] \text{ contains a cycle.} \quad (\text{MC})$$

Lemma 3.1 (Two-mark shore). *Let $A \subsetneq V(H)$, put $P = H[A]$, and suppose that P is connected and subcubic with minimum degree two. Suppose $|\delta_H(A)| \in \{2, 3\}$, the boundary edges have distinct ends in A , and exactly two marks lie in $E(P)$. If (MC) holds for every cyclic $X \subseteq A$, then one cycle of P contains both marks.*

Proof. Delete the bridges of P and form the tree of bridge-components. If P has no bridge, it is 2-edge-connected. A 2-edge-connected subcubic graph of minimum degree two has no cutvertex: two components behind a cutvertex would each require at least two incident edges at that vertex, giving degree at least four. Thus P is 2-connected. Subdivide the two marked edges. Any two vertices of a 2-connected graph lie on a common cycle, and suppressing the subdivisions gives a cycle through both marks.

Now suppose the bridge-component tree is nontrivial. Every leaf component K is cyclic; a single isolated vertex would have only one incident bridge but degree at least two in P . Let $p(K)$ count the pole-boundary edges incident with K , and let $s(K)$ count its internal marks. Exactly one bridge leaves K , so (MC) gives

$$1 + p(K) + s(K) \geq 4.$$

If K does not contain both marks, then $s(K) \leq 1$, hence $p(K) \geq 2$. Two such leaves would consume at least four distinct pole-boundary edges, impossible because there are at most three. Therefore a leaf contains both marks. That bridge-component is 2-edge-connected, subcubic, and of minimum degree two, so the preceding argument supplies the desired cycle. $\square$

Lemma 3.2 (One-mark shore). *Under the setup of Lemma 3.1, suppose instead that $|\delta_H(A)| = 3$, its ends in A are u_1, u_2, u_3 , and exactly one mark e lies in $E(P)$. If (MC) holds on every cyclic $X \subseteq A$, then for every pair $i \neq j$ there is a u_i - u_j path in P containing e .*

Proof. If P had a bridge, a leaf K of its bridge-component tree would be cyclic. With $p(K)$ and $s(K) \in \{0, 1\}$ as above,

$$1 + p(K) + s(K) \geq 4.$$

An unmarked leaf consumes all three pole-boundary edges, while a marked leaf consumes at least two. Two leaves cannot do so using only three boundary edges. Hence P is 2-edge-connected and, by the subcubic cutvertex argument, 2-connected.

Fix $i \neq j$, add an auxiliary edge $u_i u_j$, and use the standard fact that any two edges of a 2-connected graph lie on a common cycle. The assertion remains valid if the auxiliary edge is parallel to an existing edge. A cycle containing the auxiliary edge and e , with the auxiliary edge deleted, is the required path. $\square$

Corollary 3.3 (Low-cut reductions). *If a graph satisfying (MC) has no binary cycle containing all four marks with an even number of marks in each component, then it has no cyclic two-edge cut. Moreover, an unmarked cyclic three-edge cut cannot split the marks $2 + 2$.*

Proof. For a cyclic two-cut, (MC) puts at least two internal marks on each shore. Since there are four marks in total, the cut is unmarked and each shore has exactly two. Lemma 3.1 gives one two-mark cycle on each shore, whose union is a valid binary cycle. The same argument applies to the two shores of an unmarked $2 + 2$ cyclic three-cut. $\square$

4 The four-mark core theorem

Theorem 4.1 (Four-mark core closure). *Let H be a connected simple cubic graph and let S be a four-edge matching. Assume:*

- (i) *H has a Tait colouring in which every member of S has one common colour c ; and*
- (ii) *the marked cyclic-cut inequality (MC) holds.*

Then there is a binary cycle $Q \subseteq E(H)$ containing S such that every nonempty component of Q contains an even number of marks.

Remark 4.2. Universal separation is not a hypothesis of Theorem 4.1. By Lemma 2.4, it is one sufficient way to obtain hypothesis (i). The proof below uses only the one fixed common-colour Tait colouring.

Proof. Suppose that no such Q exists. By Corollary 3.3, H has no cyclic two-edge cut.

Step 1: H is 3-connected. Lemma 2.3 excludes bridges. A cutvertex in a bridgeless cubic graph is impossible, because one component behind the cutvertex would receive only one of its three incident edges, making that edge a bridge.

Suppose $\{u, v\}$ is a two-vertex cut. Every component D of $H - \{u, v\}$ has at least three boundary edges; otherwise the cubic degree sum makes both shores of $\delta_H(D)$ cyclic, contrary to the absence of a cyclic cut of size at most two. There are at least two components and at most six boundary incidences at u, v . Hence $uv \notin E(H)$, there are exactly two components D_1, D_2 , and each has three boundary edges. Neither component sends all three edges to only one of u, v , since the other would then be a cutvertex. Relabel so that D_1 sends two edges to u and one to v . Then $\delta_H(D_1 \cup \{u\})$ has size two. Both shores are cyclic by the cubic degree count and simplicity, a contradiction. Thus H is 3-connected.

Step 2: eliminate marked cyclic three-cuts. Let $B = \delta_H(A)$ be a cyclic three-edge cut and write

$$r = |B \cap S|, \quad a = |S \cap E(H[A])|, \quad d = |S \cap E(H[V(H) \setminus A])|.$$

The inequality (MC) on the two shores gives $a, d \geq 1$, and

$$a + d + r = 4. \tag{4.1}$$

In particular, $r \leq 2$. Since a shared boundary end would expose a cyclic cut of size at most two, the three cut edges have distinct ends on each shore.

If $r = 2$, then $a = d = 1$. On each shore, Lemma 3.2 gives a path through its unique internal mark between the ends of the two marked cut edges. The two paths and those two cut edges form one cycle containing all four marks.

Suppose $r = 1$. Up to exchanging shores, $a = 1$ and $d = 2$. Cap the two-mark shore by adjoining a new cubic vertex z at its three boundary ends; call the resulting cubic graph $\tilde{D}$. Lemma 2.3 shows that the fixed Tait colouring caps properly. The cap is 3-edge-connected. Indeed, it has no bridge, and a two-edge cut in a cubic graph has two cyclic shores; the shore avoiding z would induce a cyclic two-cut of H .

Let F consist of the cap edge corresponding to the marked member of B and the two internal marks. These three edges are independent. They contain no odd cut. A one-edge cut is absent. If the three edges formed a three-edge cut, that cut could not be a vertex star because F is independent. For a nontrivial three-cut, take the cyclic shore avoiding z . Both internal marks would be boundary edges and there would be no internal mark, contradicting $3 + 0 \geq 4$ in (MC). Imported result (I1) therefore gives a cycle through F .

At z , that cycle uses the marked cap edge and one other cap edge. Deleting z leaves a path through both internal marks between the corresponding boundary positions. On the one-mark shore, Lemma 3.2 supplies a path through its mark for the same boundary pair. Gluing with the two selected cut edges gives one four-mark cycle. Both cases contradict the assumed failure. Hence every cyclic three-cut that remains is unmarked.

By Corollary 3.3, an unmarked cyclic three-cut cannot split the marks $2 + 2$. Equation (4.1) shows that every such cut splits them $1 + 3$.

Step 3: organize all remaining cuts at once. Apply imported result (I3). Split along cyclic three-cuts, cap each shore by a root vertex, and continue until the factors are cyclically 4-edge-connected. Reverse reconstruction gives a tree $\mathcal{T}$: its nodes are the final factors and its edges record the vertex 3-sums. Each tree edge corresponds to a principal cyclic three-cut of H .

All principal cuts are unmarked, so every marked edge belongs to one factor. Give a node weight equal to its number of marks. Deleting any tree edge splits the total weight four as $1 + 3$. There is a node K for which every component of $\mathcal{T} - K$ has weight one. To prove this, begin at any node and move into a component of weight three whenever one exists. The walk never backtracks and must stop. At the stopping node no component has weight three; weights two and zero are excluded respectively by the $1 + 3$ rule and by positivity on both sides of every tree edge.

Let K contain r marked edges. It has $q = 4 - r$ incident one-mark branches, represented in the capped factor by root vertices $z_1, \dots, z_q$. Transfer the fixed Tait colouring to the cap. The r central marks have colour c ; at each z_j , let f_j be the unique incident edge of colour c . Let

$$F = (S \cap E(K)) \cup \{f_1, \dots, f_q\},$$

interpreted as a set, so a shared edge between adjacent cap roots is included once. The set F is a matching, $1 \leq |F| \leq 4$, it contains every central mark, and an edge of F is incident with each z_j .

If $|F| = 4$, imported result (I2) supplies a cycle through F . If $|F| \leq 3$, then F contains no odd cut: a one-cut is excluded by 3-edge-connectivity, a trivial three-cut is not a matching, and a nontrivial three-cut is excluded by cyclic 4-edge-connectivity. Imported result (I1) again supplies a cycle through F . Denote the resulting central cycle by Z .

For every j , the cycle Z passes through z_j and selects two of its three boundary positions. The corresponding entire branch of $\mathcal{T} - K$ is a shore with exactly one mark. Lemma 3.2 gives a path through that mark for the selected boundary pair. Substitute these paths while reversing the 3-sums. Each substitution preserves a single cycle. If two adjacent cap roots share the selected edge $z_i z_j$, reversing both sums turns that single edge into the single boundary edge between the two expanded shores; it is not duplicated. The final cycle contains the r central marks and one mark from each of the q branches, hence all four marks. This final contradiction proves the theorem. $\square$

Remark 4.3 (Why the conclusion allows two components). An unmarked cyclic three-cut with a $2 + 2$ mark split can be closed by one two-mark cycle on each shore. Nothing in the hypotheses forces those two cycles to merge. Thus the binary-cycle conclusion is the proved statement; an unconditional one-cycle strengthening is not claimed.

5 A forbidden root edge and the four-cut gate

We now use a stronger fixed-colouring hypothesis and obtain a rooted conclusion in a more highly connected class.

Theorem 5.1 (Independent four-cut gate). *Let H be a finite simple cyclically 4-edge-connected cubic graph. Let $S = \{s_1, s_2, s_3, s_4\}$ be a matching and let $f \notin S$. Assume that H has a Tait colouring in which*

- (i) *all members of S have one common colour c ;*
- (ii) *f has a different colour; and*
- (iii) *S is separated in this fixed colouring.*

If no cycle of $H - f$ contains all four members of S , then f belongs to an independent cyclic four-edge cut of H .

Proof. Write $f = uv$,

$$N_H(u) - \{v\} = \{u_1, u_2\}, \quad N_H(v) - \{u\} = \{v_1, v_2\},$$

and form the root-edge reduction

$$R = H \ominus f := H - \{u, v\} + \{\bar{u} = u_1 u_2, \bar{v} = v_1 v_2\}. \quad (5.1)$$

The graph H has at least eight vertices because S is a four-edge matching. It has no triangle: the triangle boundary has size three and its complementary shore is cyclic by the cubic degree count, contrary to cyclic 4-edge-connectivity. Consequently the four neighbours appearing in (5.1) do not create loops, parallel edges, or coincident new edge objects. Also H is 3-connected (for cubic graphs, cyclic connectivity and ordinary connectivity agree through order three), so deleting the adjacent pair u, v leaves a connected graph. Thus R is connected, simple, and cubic.

Map every mark $s \in S$ to an edge $\bar{s} \in E(R)$ by

$$\bar{s} = \begin{cases} \bar{u}, & \text{if } s \text{ is incident with } u, \\ \bar{v}, & \text{if } s \text{ is incident with } v, \\ s, & \text{otherwise.} \end{cases} \quad (5.2)$$

The four images are distinct. We claim that they form a matching. Only adjacency between a new image and an unchanged image needs proof. Suppose, without loss of generality, that $s = uu_1$ is a mark of colour c , while f has colour a . Then uu_2 has the third colour b . An unchanged mark incident with u_2 would lie with s on the bc -factor cycle, contradicting fixed-colouring separation. A mark incident with u_1 would already be adjacent to s . The other cases are symmetric.

If R were cyclically 4-edge-connected, imported result (I2) would give a cycle through the four independent images. Replace $\bar{u}$, when used, by u_1uu_2 , and similarly replace $\bar{v}$. The lifted cycle lies in $H - f$ and contains every original mark, contrary to hypothesis. Hence R has a cyclic cut $\delta_R(X)$ of size $k \leq 3$.

We pull this cut back to H . If u_1, u_2 lie on opposite shores, then $\bar{u}$ is a cut edge; placing u on either shore replaces $\bar{u}$ by exactly one crossing edge. If u_1, u_2 lie on one shore, place u there and create no crossing edge. The same choices apply to v . Unless both neighbour pairs are unsplit and lie on opposite shores, choose u, v on the same shore. Then f is internal, the cut size remains $k \leq 3$, and both shore cycles lift: an internal new edge is replaced by its two-edge path, while a crossing new edge was not part of either shore cycle. This contradicts cyclic 4-edge-connectivity of H .

Therefore u_1, u_2 lie together on one shore and v_1, v_2 lie together on the other. Restoring u, v on those forced shores gives

$$\delta_H(X^\uparrow) = \delta_R(X) \cup \{f\}. \quad (5.3)$$

Both shores remain cyclic, so $k + 1 \geq 4$. Since $k \leq 3$, we have $k = 3$.

It remains to prove independence. If two edges of the cyclic three-cut $\delta_R(X)$ shared an endpoint w on one shore, then only one edge at w would remain internal. No cycle of that shore uses w . Moving w to the other shore replaces two crossing edges by one and preserves cycles on both shores, producing a cyclic two-cut of R . The pullback argument just given excludes such a cut, since it would produce a cyclic cut of size at most three in H . Thus the three edges of $\delta_R(X)$ are independent. In the forced configuration they avoid the new edges and hence avoid u, v . Together with f , they form an independent cyclic four-edge cut of H . $\square$

Corollary 5.2 (Rooted closure at cyclic connectivity five). *Under the hypotheses of Theorem 5.1, if H is cyclically 5-edge-connected, then $H - f$ has a cycle containing all four marks.*

Proof. Otherwise Theorem 5.1 produces a cyclic four-edge cut. $\square$

Remark 5.3 (What remains at cyclic connectivity four). Theorem 5.1 is a one-way obstruction localization, not a characterization: a root-containing independent four-cut need not prevent an avoiding cycle. The unresolved case is the corresponding four-pole interface. No unrestricted rooted theorem follows from the gate alone.

6 Cycle translations in an exceptional rooted interface

This section records a second, algebraic restriction on a rooted interface. Its “root” is a distinguished cap edge whose possible D_5 -labels are studied; it is not the forbidden edge of Theorem 5.1.

The conclusion is a necessary obstruction, not a proof that the exceptional interface is impossible. Throughout this section, $\mathcal{R}_5$ and $\mathcal{C}_5$ denote exact signatures inside one fixed five-coordinate D_5 universe. This is narrower than the published unbounded-colour relation.

6.1 The D_5 model

Recall Γ , q , and $D_5 = q^{-1}(1)$ from (2.1)–(2.2). We identify a member of D_5 with its weight-two incidence vector and write addition in Γ as symmetric difference. A D_5 -labelling of a cubic multipole assigns a member of D_5 to every proper edge and semiedge so that the three incident labels sum to zero at each internal vertex. Such a local triple is exactly the three-edge set of a triangle in K_5 .

Let Q be a cubic three-pole whose ordered connector word has been normalized to

$$\tau = (01, 02, 12), \quad (6.1)$$

and let r be a distinguished proper edge. Its exact root signature is

$$\mathcal{R}_5(Q, r) = \{A \in D_5 : Q \text{ has a } D_5\text{-labelling with word } \tau \text{ and root label } A\}. \quad (6.2)$$

The stabilizer of the ordered word τ fixes 0, 1, 2 and may swap 3, 4. Therefore $\mathcal{R}_5(Q, r)$ is a union of the seven orbits

$$\{01\}, \{02\}, \{12\}, \{03, 04\}, \{13, 14\}, \{23, 24\}, \{34\}. \quad (6.3)$$

Lemma 6.1 (Cycle translation). *Let λ be a normalized root labelling of Q , let C be a cycle of proper edges, and let $h \in \Gamma$. If*

$$\lambda(e) + h \in D_5 \quad (e \in E(C)), \quad (6.4)$$

then adding h to every label on C , and changing no other label, gives another normalized root labelling.

Proof. Every vertex of C is incident with exactly two translated edges, so the change in its incident-label sum is $h + h = 0$. All other vertices are unchanged. A proper cycle contains no connector semiedge, so the word τ in (6.1) remains fixed. Condition (6.4) says that all translated values remain in the allowed label set. $\square$

Lemma 6.2 (Translation blockers). *Let $L \subseteq E(K_5) = D_5$ be nonempty, with its edges connected under edge adjacency. The only $h \in \Gamma$ for which*

$$A + h \in D_5 \quad (A \in L) \quad (6.5)$$

is $h = 0$ if and only if the graph $(\{0, 1, 2, 3, 4\}, L)$

(i) spans all five coordinate vertices; and

(ii) is either non-bipartite or is the four-edge star $K_{1,4}$.

Proof. Every nonzero element of Γ has weight two or four. First let h have weight four, say $h = \{0, 1, 2, 3, 4\} \setminus \{v\}$. For $A \in D_5$, the symmetric difference $A + h$ has weight two exactly when A avoids v . Thus all weight-four translations are blocked exactly when L spans all five vertices.

Now let $h = B$ have weight two. For $A \in D_5$,

$$|A \triangle B| = 2 \iff |A \cap B| = 1.$$

Consequently B translates every member of L within D_5 exactly when every edge of L crosses the cut

$$B \mid (\{0, 1, 2, 3, 4\} \setminus B). \quad (6.6)$$

If L spans and is non-bipartite, it is contained in no such cut. If L is spanning and bipartite, edge-adjacency connectedness makes it connected, so its bipartition is unique. A $2 + 3$ bipartition gives a weight-two translation. A $1 + 4$ bipartition has no side of size two; connectedness and spanning force all four incident edges, namely $K_{1,4}$. The weight-two and weight-four cases prove the equivalence. $\square$

Remark 6.3. Up to coordinate permutation, the inclusion-minimal connected blockers are $K_{1,4}$, the bull graph, and C_5 . This follows directly: a minimal spanning non-bipartite graph contains a triangle or a five-cycle. Minimality gives C_5 in the latter case. Starting from a triangle, the unused vertices must attach at distinct triangle vertices: attaching both to one vertex contains a star blocker, while a pendant path leaves a triangle plus a disjoint edge after its joining edge is deleted. The remaining case is the bull.

Theorem 6.4 (Cycle-support obstruction). *Suppose*

$$\mathcal{R}_5(Q, r) = \{A\}, \quad A \in \{01, 02, 12, 34\},$$

and r is a nonbridge of the proper graph of Q . In every normalized root labelling and on every proper cycle C containing r , the set of distinct labels

$$L_C = \{\lambda(e) : e \in E(C)\}$$

spans all five coordinates and is either non-bipartite or $K_{1,4}$.

Proof. Consecutive cycle labels share one coordinate because they are two edges of the local label triangle. Thus L_C is nonempty and connected under edge adjacency. If it were not a blocker, Lemma 6.2 would give a nonzero h satisfying (6.4). By Lemma 6.1, translation on C gives another normalized labelling whose root label is $A + h \in D_5$. Since $h \neq 0$, this differs from A , contradicting the singleton signature. Lemma 6.2 gives the stated classification. $\square$

6.2 Exact factorization across the cyclic three-cut

We now state the precise setting in which the singleton arises. Let $G = P + \{e, f\}$ be a bridgeless cubic cap with no cyclic two-edge cut. Assume that a cyclic three-edge cut, disjoint from e, f , has connected shores, splits the four terminals of P as $2 + 2$, and contains one whole cap edge on each shore. Cutting the three cut edges gives rooted three-poles (Q_1, e) and (Q_2, f) . Order corresponding connectors, normalize them to τ , and write

$$R = \mathcal{R}_5(Q_1, e), \quad S = \mathcal{R}_5(Q_2, f).$$

For nonempty $R, S \subseteq D_5$, let

$$\text{rel}(R, S) \subseteq \{E, I, D\}$$

record which of equality, one-point intersection, and disjointness occur among cross-pairs $(A, B) \in R \times S$. Let π denote the pairing of the four boundary semiedges induced by the deleted cap edges. In the standard four-pole notation, a doubled boundary word (A, A, B, B) has type

$$AA, \quad AT_\pi, \quad \text{or} \quad T_\pi T_\pi$$

according as $A = B$, $|A \cap B| = 1$, or $A \cap B = \emptyset$.

Proposition 6.5 (Exact rooted factorization). *The doubled part of the fixed-five boundary signature $\mathcal{C}_5(P)$ is*

$$\begin{aligned} \mathcal{C}_5(P) \cap \{AA, AT_\pi, T_\pi T_\pi\} = & \{AA : E \in \text{rel}(R, S)\} \\ & \cup \{AT_\pi : I \in \text{rel}(R, S)\} \\ & \cup \{T_\pi T_\pi : D \in \text{rel}(R, S)\}. \end{aligned} \quad (6.7)$$

Proof. Given a labelling of P with doubled word (A, A, B, B) , restore e with label A and f with label B . Cutting the three-edge cut gives two rooted labellings with a common ordered connector triangle. One global coordinate permutation normalizes it to τ , placing the transformed root labels in R, S ; equality, intersection size, and disjointness are invariant under that permutation.

Conversely, take defining labellings for any $A \in R$ and $B \in S$. Their connector words are identically τ , so corresponding semiedges glue. Delete the two roots while retaining their labels on the four boundary semiedges. The resulting word is (A, A, B, B) , with the type determined by the cross-relation. Both inclusions follow. $\square$

Lemma 6.6 (Sparse cross-relations). *Let R, S be nonempty and invariant under the stabilizer of τ .*

(a) $\text{rel}(R, S) = \{E\}$ if and only if

$$R = S = \{A\} \quad \text{for some } A \in \{01, 02, 12, 34\}. \quad (6.8)$$

(b) $\text{rel}(R, S) = \{D\}$ if and only if, after possibly swapping R, S ,

$$\emptyset \neq R \subseteq \{01, 02, 12\}, \quad \emptyset \neq S \subseteq \{d \in D_5 : d \cap a = \emptyset \text{ for every } a \in R\}, \quad (6.9)$$

where S is stabilizer-invariant. There are thirteen unordered patterns in (6.9).

Proof. If every cross-pair is equal, choosing one member on either side forces both sets to be the same singleton. The singleton is stabilizer-invariant exactly for the four fixed labels in (6.8), proving (a).

For (b), every pair described in (6.9) is disjoint. Conversely, suppose every cross-pair is disjoint. If one side contains 34, the other side consists of triangle edges. If one side contains a mixed edge $u3$, invariance also puts $u4$ there, and the only edge disjoint from both is the triangle edge on $\{0, 1, 2\} \setminus \{u\}$. Thus, after swapping, one side is a nonempty subset of $\{01, 02, 12\}$, and the other is exactly restricted as in (6.9). For a one-edge triangle subset the common disjoint neighbourhood has two stabilizer orbits, giving three nonempty invariant choices; this contributes $3 \cdot 3 = 9$. A two-edge subset has neighbourhood $\{34\}$, contributing three, and the three-edge subset contributes one. Hence $9 + 3 + 1 = 13$. $\square$

Lemma 6.7 (Cap roots are nonbridges). *In the cap geometry preceding Proposition 6.5, neither e nor f is a bridge of its rooted shore.*

Proof. Suppose e is a bridge of Q_1 , splitting its proper graph into connected vertex sets X, Y . Let k of the three cut connectors be incident with X . If $k = 0$ or 3 , then e is a bridge of G , contrary to bridgelessness. Thus $k \in \{1, 2\}$. Choose $Z = X$ for $k = 1$, and $Z = Y$ for $k = 2$. Exactly two edges of G leave Z : e and one connector.

If Z has n vertices and m internal edges, cubic degree counting gives $3n = 2m + 2$, so

$$m - n + 1 = n/2 > 0.$$

Hence Z contains a cycle. Its complement is connected: the other bridge component of Q_1 connects through the two remaining connectors to the connected opposite shore Q_2 . Applying the same degree count to the complement gives a cycle there as well. The two leaving edges form a cyclic two-edge cut of G , a contradiction. The argument for f is symmetric. $\square$

Máčajová, Mazzuoccolo, and Tabarelli [3] isolate the following two exceptional type sets in their unbounded-colour signature relation:

$$\begin{aligned} \mathcal{E}_4(i; j, k) &= \{AA, AT_j, AT_k, T_j T_k\}, \\ \mathcal{E}_5(i; j, k) &= \{T_i T_i, T_j T_j, T_k T_k, T_i T_j, T_i T_k\}, \quad \{i, j, k\} = \{2, 3, 4\}. \end{aligned} \tag{6.10}$$

Below, the statements “ $\mathcal{C}_5(P) = \mathcal{E}_4$ ” and “ $\mathcal{C}_5(P) = \mathcal{E}_5$ ” use the same type masks as conditional targets in the fixed-five model. They do not identify $\mathcal{C}_5(P)$ with the full published relation. Intersecting these displayed sets with the three types in Proposition 6.5 gives the following literal consequence.

Corollary 6.8 (Exceptional rooted obligations). *(i) If $\mathcal{C}_5(P) = \mathcal{E}_4(i; j, k)$ and $\pi = i$, then $\text{rel}(R, S) = \{\mathbf{E}\}$. Thus $R = S = \{A\}$ for one fixed label $A \in \{01, 02, 12, 34\}$, both roots are nonbridges, and every cycle through either root has a support classified by Theorem 6.4.*

(ii) If $\mathcal{C}_5(P) = \mathcal{E}_4(i; j, k)$ and $\pi \in \{j, k\}$, then $\text{rel}(R, S) = \{\mathbf{E}, \mathbf{I}\}$.

(iii) If $\mathcal{C}_5(P) = \mathcal{E}_5(i; j, k)$, then the cross-relations are disjointness only:

$$\text{rel}(R, S) = \{\mathbf{D}\}.$$

Hence one of the thirteen patterns in Lemma 6.6(b) occurs.

Proof. For $\mathcal{E}_4$, the intersection with $\{AA, AT_\pi, T_\pi T_\pi\}$ is $\{AA\}$ when $\pi = i$, and $\{AA, AT_\pi\}$ when $\pi = j$ or k . For $\mathcal{E}_5$, it is $\{T_\pi T_\pi\}$ for every π . Proposition 6.5, Lemmas 6.6 and 6.7, and Theorem 6.4 give the assertions. $\square$

6.3 Missing base pairs force proper-core cuts

For a normalized labelling ϕ and a coordinate a , let

$$E_{\bar{a}}(\phi) = \{e \text{ a proper edge of } Q : a \notin \phi(e)\}, \quad h_a = \{0, 1, 2, 3, 4\} \setminus \{a\}.$$

Lemma 6.9 (Complement-cycle switch). *Suppose $a \notin \phi(r)$. If r lies on a proper cycle $C \subseteq E_{\bar{a}}(\phi)$, then replacing every label $\phi(e)$, $e \in C$, by $\phi(e) \triangle h_a$ gives another normalized labelling. Consequently, if $\phi(r) \triangle h_a \notin \mathcal{R}_5(Q, r)$, then r is a bridge of $E_{\bar{a}}(\phi)$.*

Proof. Every label on C is a two-subset of the four-set h_a ; symmetric difference with h_a is its complementary two-subset and therefore remains in D_5 . At a cycle vertex the same vector h_a is added twice, so the vertex equation is unchanged. No boundary semiedge is changed. The last assertion uses the elementary fact that an edge of a finite graph is a nonbridge exactly when it lies on a cycle. $\square$

Define the three base pairs

$$\mathcal{P}_0 = \{12, 03, 04\}, \quad \mathcal{P}_1 = \{02, 13, 14\}, \quad \mathcal{P}_2 = \{01, 23, 24\}.$$

Theorem 6.10 (Base-pair cut certificate). *Suppose $\mathcal{R}_5(Q, r)$ is nonempty, contains a label other than 34, and contains none of $\mathcal{P}_0, \mathcal{P}_1, \mathcal{P}_2$ as a subset. Then some realizing labelling ϕ and $a \in \{3, 4\}$ make r a bridge of $E_{\bar{a}}(\phi)$.*

More precisely, an inner root label 12, 02, or 01 forces this bridge for both $a = 3$ and $a = 4$; an outer orbit without its inner mate supplies one bridge for each coordinate after applying the symmetry $3 \leftrightarrow 4$.

Proof. The fixed boundary word $(01, 02, 12)$ is invariant under $3 \leftrightarrow 4$, so the exact root signature is invariant as well. Complementation in the four-set avoiding 4 pairs

$$12 \leftrightarrow 03, \quad 02 \leftrightarrow 13, \quad 01 \leftrightarrow 23,$$

and complementation in the four-set avoiding 3 gives the same three pairs with 3, 4 interchanged. If an inner label occurs, failure of its base pair and signature invariance exclude both outer mates. If an outer orbit occurs, failure of its base pair excludes the inner mate. Lemma 6.9 gives the claimed bridges. $\square$

Here is the promised ordinary cut certificate. Let $\delta_Q^p(X)$ denote the edge cut in the proper core of Q , explicitly excluding its three boundary semiedges. If X is one component of $E_{\bar{a}}(\phi) - r$, then

$$\delta_Q^p(X) = \{r\} \dot{\cup} F_a, \quad F_a \subseteq \{e : a \in \phi(e)\}.$$

For $a = 3, 4$, no boundary label uses a . The coordinate- a support on proper edges is therefore Eulerian, so $|F_a|$ is even. If r is a nonbridge of the unlabelled proper core, then $F_a \neq \emptyset$. Thus Theorem 6.10 supplies such a nonbridge root with an odd proper-core cut of size at least three, containing r , whose other edges carry one common outer coordinate.

For completeness, if $\mathcal{R}_5(Q, r) = \{34\}$, complementation in the four-sets avoiding 0, 1, 2 shows that r is simultaneously a bridge of $E_{\bar{0}}(\phi)$, $E_{\bar{1}}(\phi)$, and $E_{\bar{2}}(\phi)$. Boundary semiedges use these coordinates, so the even- $|F_a|$ conclusion is not asserted in this case.

Remark 6.11 (Exact limitation). Corollary 6.8 neither realizes nor excludes any of the remaining root signatures. In particular, Theorem 6.4 does not prove that a nonbridge root has two possible labels: it only classifies what every root cycle must look like if the label remains unique. The cut certificate above is likewise necessary, not contradictory. The $\{E, I\}$ and thirteen disjoint cases receive no further elimination here.

6.4 Tait-cap base-pair closure

Adjoin one new vertex to the three connector ends of a rooted three-pole Q , producing a cubic graph $\hat{Q}$.

Theorem 6.12 (Tait-cap closure). *Let (Q, r) be normalized by the connector word τ in (6.1). If $\widehat{Q}$ is Tait-colourable and r is a nonbridge of the proper graph of Q , then*

$$\mathcal{P}_i \subseteq \mathcal{R}_5(Q, r) \quad \text{for some } i \in \{0, 1, 2\}. \quad (6.11)$$

Proof. Fix a Tait colouring of $\widehat{Q}$. The three cap edges have distinct colours. Map their colours, in physical connector order, to

$$01, \quad 02, \quad 12.$$

Give every edge of $\widehat{Q}$ the D_5 -label assigned to its colour. Every cubic vertex sees all three labels, whose symmetric difference is zero. Deleting the cap vertex therefore gives a normalized root labelling of Q .

The root r receives one of 01, 02, 12. Since it is a nonbridge, the elementary bridge-cycle equivalence gives a cycle C of proper edges containing r . Translate every label on C , first by

$$h_3 = 0123$$

and, in a separate copy of the original labelling, by

$$h_4 = 0124.$$

The only labels initially occurring on C are 01, 02, 12, and

$$\begin{array}{c|ccc} & 01 & 02 & 12 \\ \hline + h_3 & 23 & 13 & 03 \\ + h_4 & 24 & 14 & 04. \end{array} \quad (6.12)$$

All six outputs remain in D_5 . At a cycle vertex the same translation is added on two incident edge-ends, so parity is unchanged; the proper cycle contains no connector semiedge. Thus both translations are normalized root labellings by Lemma 6.1. According as the original root label is 12, 02, or 01, its three displayed values are exactly $\mathcal{P}_0, \mathcal{P}_1$, or $\mathcal{P}_2$. $\square$

The one-sided exceptional consequence requires care. Let

$$\begin{array}{c|ccccccc} & \{01\} & \{02\} & \{12\} & \{03, 04\} & \{13, 14\} & \{23, 24\} & \{34\} \\ \hline \mathcal{P}_0 & I & I & DE & DEI & DI & DI & DI \\ \mathcal{P}_1 & I & DE & I & DI & DEI & DI & DI \\ \mathcal{P}_2 & DE & I & I & DI & DI & DEI & DI \end{array} \quad (6.13)$$

record the cross-relations between a base pair and each stabilizer orbit; concatenated letters denote sets of relations.

Corollary 6.13 (Corrected Tait-shore consequence). *In the cap geometry of Proposition 6.5, assume that R and S are nonempty:*

(i) *if either $\widehat{Q}_1$ or $\widehat{Q}_2$ is Tait-colourable, then*

$$\text{rel}(R, S) \notin \{\{E\}, \{D\}\};$$

(ii) *if both shore caps are Tait-colourable, then*

$$\text{rel}(R, S) \notin \{\{E\}, \{E, I\}, \{D\}\}.$$

Consequently one Tait shore excludes $\mathcal{E}_5$ and the equality-only orientation of $\mathcal{E}_4$; two Tait shores exclude both exceptional fixed-five signatures.

Proof. Suppose first that $\widehat{Q}_1$ is Tait-colourable. Lemma 6.7 and Theorem 6.12 give $\mathcal{P}_i \subseteq R$. The other signature S is nonempty and is a union of columns in (6.13). Every nonempty union contains a relation other than E, and a relation other than D. Since $\text{rel}(\mathcal{P}_i, S) \subseteq \text{rel}(R, S)$, equality-only and disjointness-only are impossible.

If both caps are Tait-colourable, there are base pairs $\mathcal{P}_i \subseteq R$ and $\mathcal{P}_j \subseteq S$. For $i = j = 0$, the pairs 12, 03 and 03, 04 exhibit D and I, respectively; coordinate permutations give the other diagonal cases. For $i = 0, j = 1$, the pairs 12, 02 and 03, 14 exhibit I and D; cyclic permutations of 0, 1, 2, and swapping the factors, give all off-diagonal cases. Hence both I and D occur in $\text{rel}(R, S)$, excluding all three sparse exceptional sets. Corollary 6.8 identifies those sets with the stated exceptional orientations. $\square$

Remark 6.14 (The mixed relation survives one-sidedly). The stronger one-sided claim is false. The invariant sets

$$R = \mathcal{P}_0 \cup \{01\}, \quad S = \{01\}$$

satisfy $\mathcal{P}_0 \subseteq R$ but

$$\text{rel}(R, S) = \{E, I\}. \quad (6.14)$$

More precisely, put

$$U_0 = \{01, 02\}, \quad U_1 = \{01, 12\}, \quad U_2 = \{02, 12\}.$$

If $\mathcal{P}_i \subseteq R$ and the cross-relation is exactly $\{E, I\}$, Table (6.13) forces

$$\emptyset \neq S \subseteq U_i, \quad R \cap S \neq \emptyset. \quad (6.15)$$

These are necessary restrictions, not an exclusion.

Corollary 6.15 (Conditional three-sum path consequence). *Suppose a 3-connected simple cap $G = P + \{e, f\}$ satisfies the cyclic-three factorization hypotheses above and has a cubic three-sum factor tree in which every principal cyclic three-edge cut separates e from f . Then the factor tree is a path between the factors containing e and f .*

If $\mathcal{C}_5(P) = \mathcal{E}_5$, both endpoint factors are non-Tait. If $\mathcal{C}_5(P) = \mathcal{E}_4(i; j, k)$, an endpoint factor is non-Tait whenever its adjacent cut has pairing i ; a Tait endpoint is possible only for pairing j or k , and then the opposite root signature satisfies (6.15). If the tree has one node, that node is the non-Tait cap supplied by Theorem 7.2.

Proof. Let u, v be the factor-tree nodes containing e, f . In a tree, the edges whose deletion separates u and v are exactly the edges of the u - v path. By hypothesis every tree edge has this property, so there is no off-path edge and the tree is a path.

For a nontrivial path, the shore cap at an end cut is precisely the endpoint factor. Corollary 6.13 excludes a Tait endpoint when the corresponding relation is $\{D\}$ or $\{E\}$. Corollary 6.8 gives disjointness-only for $\mathcal{E}_5$, equality-only for the $\pi = i$ orientation of $\mathcal{E}_4$, and the mixed relation for $\pi \in \{j, k\}$. The last case is restricted by (6.15), as claimed. $\square$

Remark 6.16 (Premise supplied by the minimal simple-cap fork). In the vertex-minimal, two-cut-reduced simple-cap fork, the elementary cut analysis shows that every nontrivial cyclic three-edge cut separates e from f : cuts meeting one or both cap edges leave a forbidden proper one- or

two-edge cut, while a terminal-free shore of a cut disjoint from the cap edges can be replaced by one cubic vertex without changing the exact fixed-five signature. The published decomposition theorem imported as (I3) assumes 3-connectivity, and that premise follows here as well. The cap is simple, bridgeless, cubic, and has no cyclic two-edge cut. A cutvertex in a bridgeless cubic graph would leave one component attached by a single edge. If $\{u, v\}$ were a two-vertex cut, every component of $G - \{u, v\}$ would have at least three boundary edges; a smaller boundary is a bridge or, by cubic degree counting and simplicity, a cyclic two-edge cut. The six incidences at u, v then force $uv \notin E(G)$, exactly two components with three boundary edges each, and a $2+1$ attachment to u, v . Adding to one component the vertex receiving two attachments produces a cyclic two-edge cut, a contradiction. Hence the minimal simple cap is 3-connected and Corollary 6.15 applies without an extra connectivity assumption.

6.5 Fork transport through an end factor

For five distinct coordinates p, q, r, s, t , define

$$Q(p; q, r; s, t) = \{qr, ps, pt\} \subseteq D_5. \quad (6.16)$$

We call every set of this form a *fork*. The three base pairs

$$\mathcal{P}_0 = \{12, 03, 04\}, \quad \mathcal{P}_1 = \{02, 13, 14\}, \quad \mathcal{P}_2 = \{01, 23, 24\} \quad (6.17)$$

are forks, and coordinate permutations preserve the class of forks.

Theorem 6.17 (Endpoint-fork lift). *Let a simple 3-connected cubic graph H be reconstructed as a path of vertex three-sums*

$$F_1, F_2, \dots, F_k.$$

Suppose a distinguished proper root edge r lies in F_1 , while the deleted vertex supplying the three external connectors lies in F_k . Let w be the virtual vertex of F_1 at its first principal cut. Normalize the ordered external connector word to $\tau = (01, 02, 12)$.

If the normalized whole-shore root signature $R(H, r)$ is nonempty and the endpoint signature $R(F_1, w, r)$ contains one of the base pairs in (6.17), then $R(H, r)$ contains a fork.

Proof. Choose one labelling of H . The labels on the first principal three-cut form an ordered triangle $\alpha = (a, b, c)$: their xor is zero and each lies in D_5 . There is a coordinate permutation $\sigma \in S_5$ taking α , in its physical order, to τ , since both words are ordered edge sets of triangles in K_5 .

Restrict the chosen labelling to the fixed remainder $F_2, \dots, F_k$, retaining α at its F_1 -interface and the already normalized external word at the far end. If $\mathcal{P}_i \subseteq R(F_1, w, r)$, choose the three endpoint labellings with connector word τ and root values in $\mathcal{P}_i$. Apply the same permutation σ^{-1} to all three endpoint labellings. Their connector word becomes α , so every one glues to the same fixed remainder. The resulting whole shore root values are exactly $\sigma^{-1}(\mathcal{P}_i)$, which is a fork. $\square$

Notice that nonemptiness also transports in the needed direction: restricting one whole-shore labelling to F_1 and normalizing its interface triangle proves that the endpoint signature is nonempty. In the application below the endpoint root is a nonbridge. A cyclically four-edge-connected simple cubic factor (and K_4) is 3-connected; deleting its virtual vertex leaves a 2-connected graph.

Lemma 6.18 (Fork crossing). *If nonempty $R, S \subseteq D_5$ each contain a fork, then*

$$\{I, D\} \subseteq \text{rel}(R, S). \quad (6.18)$$

Moreover, one fork-containing side and any nonempty opposite side cannot have an equality-only or a disjointness-only relation.

Proof. It suffices first to compare forks $P \subseteq R$ and $Q \subseteq S$. Write $P = \{qr, ps, pt\}$.

Suppose no member of Q is disjoint from a member of P . A two-set meeting both ps and pt either contains p or is st ; the latter misses qr . In the former case, also meeting qr leaves only pq or pr . Thus the three distinct members of the fork Q would lie in a two-element set, a contradiction. Hence a disjoint cross-pair exists.

Now suppose there is no unequal intersecting cross-pair. Neither ps nor pt can belong to Q , since they intersect one another unequally. A label not equal to a member of P would have to be disjoint from all three members of P , but their union contains all five coordinates. Every member of Q would therefore equal qr , again contradicting that a fork has three distinct members. This proves (6.18).

For the one-sided statements, equality-only would force both sides to be the same singleton, while a fork has three members. Disjointness-only would require an opposite label disjoint from all three fork members, which is impossible by the preceding all-five-coordinate observation. $\square$

The $30^2 = 900$ ordered fork pairs split into 330 with cross-relation $\{I, D\}$ and 570 with $\{E, I, D\}$. This literal table is only an error check; the displayed proof is independent of it.

7 Fixed-five rooted finite frontiers

7.1 A human-checkable cap reduction

Let P be a connected simple proper core with four specified terminals $T = \{t_1, t_2, t_3, t_4\}$, each of proper degree two, and all other vertices of proper degree three. Attaching one semiedge at each terminal gives a terminal-distinct cubic four-pole. A perfect matching of T is *simple* for P if neither of its pairs is already an edge of P .

Lemma 7.1 (Simple cap matching). *At least one of the three perfect matchings of T is simple for P .*

Proof. Suppose each perfect matching contains an edge of $P[T]$, and choose one such edge from each of the three matching classes of K_4 . The three chosen edges form either a three-edge star or a triangle. A star would give its terminal centre three proper neighbours, although that terminal has degree two. In the triangle case, all three triangle vertices have both proper incident edges inside the triangle, making it a component of P disjoint from the fourth terminal. Both possibilities contradict the setup. $\square$

The non-Tait consequence used next retains the fixed-five scope. Máčajová, Mazzuoccolo, and Tabarelli [3, Lemma 3.8] prove that a Tait-colourable pole has no isolated vertex in its signature graph. The T_i vertex is isolated in $\mathcal{E}_4$, and A is isolated in $\mathcal{E}_5$. Moreover, the cited construction uses at most four colours, so the same exclusion holds when the exact signature is restricted to one fixed five-set.

Theorem 7.2 (Simple-cap enumeration reduction). *Suppose no proper edge of P is a bridge and $\mathcal{C}_5(P)$ is one of the fixed-five target masks $\mathcal{E}_4, \mathcal{E}_5$ in (6.10). Some simple matching on T adds two independent edges e, f to produce a connected simple bridgeless cubic graph $G = P + \{e, f\}$ which is not Tait-colourable. Deleting e, f recovers P .*

Proof. Choose the matching by Lemma 7.1. Its pairs are nonedges with four distinct endpoints, so adding them preserves simplicity and makes every terminal cubic. The graph remains connected. Every old edge lies on a cycle because the proper core has no bridge. Each new edge lies on a cycle because its endpoints are joined by a path in connected P . Thus G is bridgeless.

If G had a Tait colouring, deleting e, f and retaining their colours on the four resulting semiedges would Tait-colour P , contrary to the fixed-five consequence of the cited lemma. $\square$

7.2 Root base pairs through order 17

A rooted three-pole core of order n here is a connected simple graph with exactly three degree-two terminals, all other vertices cubic, and one distinguished proper edge. Its three boundary semiedges are ordered and normalized to (01, 02, 12).

Theorem 7.3 (Finite rooted base-pair theorem). *For every odd $n \leq 17$, every nonempty fixed-five root signature of a nonbridge root in such an order- n core contains at least one of $\mathcal{P}_0, \mathcal{P}_1, \mathcal{P}_2$.*

This is a computer-assisted finite theorem. Canonical input was generated by

```
geng -cq -d2 -D3 n m:m,      m = (3n - 3)/2,
```

which forces exactly three degree-two vertices. At order 17, eight canonical shards contain 654 676 cores and 15 645 623 nonbridge roots. Two separately written engines were used: a direct finite-domain propagator and an incremental SAT encoding with a one-hot ten-label domain and the complete local support table. Each reports

empty signatures	337 059,
signatures containing a base pair	15 308 564,
violations	0,
solver calls	52 216 251.

Their deterministic transcript digests agree separately on all eight shards. The retained verifier checks generator status, row accounting, both implementation summaries, and every digest equality. Lower odd orders are included in the archived report.

Theorem 7.3 is not an induction and supplies no universal root-signature theorem. Combined with Theorem 6.10, it says that the ordinary cut certificate does not occur among nonempty nonbridge roots in this finite simple-core range because a base pair is always present.

7.3 Endpoint factors through order 26

Theorem 7.4 (Finite endpoint base-pair theorem). *Let F be a cyclically 4-edge-connected simple cubic graph of even order at most 26. Delete any vertex z , order and normalize its three former incident edges to (01, 02, 12), and distinguish any nonbridge proper edge r of $F - z$. If the resulting fixed-five root signature is nonempty, it contains one of the three base pairs in (6.17).*

For factor order at most 18 this is Theorem 7.3, applied to the vertex-deleted core. At factor orders 20, 22, 24, 26, a Tait-colourable F is covered constructively by Theorem 6.12. The remaining complete cyclically four-edge-connected non-Tait sources contain respectively

$$6, \quad 31, \quad 155, \quad 1297$$

graphs. Deleting every vertex gives 38 244 rooted cores. Independent incremental-CaDiCaL and direct finite-domain classifiers each test 1 360 452 nonbridge roots in 4 157 844 adaptive solver calls. They report no empty row and no violation. Their complete uncompressed transcripts are byte-identical, with SHA-256

$$\text{bfb90fd89b43e66f02ada87abcb87ab643d87a4b36dba616cd1f3e5d603b7364}.$$

An independently written verifier reconstructs the sources and all vertex deletions, checks the graph premises and every nonbridge root, and reproduces the report. Source completeness for orders 20, 22, 24, 26 relies on Snarkhunter 2.0b and its documented option semantics.

7.4 The cyclically four-connected order-22 cap corpus

Theorem 7.5 (Finite full-signature cap census). *Let G be a cyclically 4-edge-connected, bridgeless, non-Tait, connected simple cubic graph on 22 vertices, and let e, f be independent edges. Then the deletion pole $G - \{e, f\}$ has all ten parity-permitted boundary types in its fixed-five signature:*

$$\mathcal{C}_5(G - \{e, f\}) \text{ has mask } 0\mathbf{x}3ff.$$

The quantifier is finite and exhaustive. Of 7 319 447 canonical connected simple cubic graphs on 22 vertices, 7 187 627 are bridgeless. Independent direct edge-colouring and perfect-matching filters agree on exactly 12 892 non-Tait graphs and on the hard corpus digest

$$230f2cd88e72011d73a40c5f3d7d5f9fb0fca6110de2ec2541581bc50982037c.$$

Exact cut classification leaves 31 cyclically 4-edge-connected graphs. Deleting every independent edge pair gives 14 322 ordered pole records.

Two classifiers ask ten questions for each record. One uses CaDiCaL; the other is independently written finite-domain code. Both return

$$\begin{aligned} \text{rows} &= 14\,322, \\ \text{queries} &= \text{SAT answers} = 143\,220, \\ \text{nonfull rows} &= 0. \end{aligned}$$

Their uncompressed output streams are byte-identical, with SHA-256

$$52edd8a189012acefd3565a79d5c58b6f209bbe40156560eb57e2c569fd7c091.$$

These checks prove Theorem 7.5 conditional on the generator and two implementations.

Corollary 7.6 (Complete order-22 cap consequence). *No bridge-free connected simple terminal-distinct order-22 four-pole has fixed-five signature $\mathcal{E}_4$ or $\mathcal{E}_5$.*

Proof. Theorem 7.2 places some simple cap in the complete corpus of 12 892 bridgeless non-Tait order-22 graphs. Deleting all independent edge pairs gives 5 956 104 pole rows. Independent CaDiCaL and direct finite-domain classifiers make 23 747 129 exact signature queries per implementation and agree that none of the rows is exceptional. The complete replay reconstructs the canonical source, every deletion row, every shard digest, and both implementation summaries. $\square$

7.5 Cyclically four-connected caps through order 26

Theorem 7.7 (Finite cap classifications at orders 24 and 26). *Let G be a cyclically 4-edge-connected, bridgeless, non-Tait, connected simple cubic graph of order 24 or 26, and let e, f be independent edges. Then*

$$\mathcal{C}_5(G - \{e, f\}) \text{ has mask } 0x3ff.$$

At order 24 the retained Snarkhunter source has 155 caps. Its 86 490 independent-edge deletions require 864 900 exact boundary queries per implementation; every query is satisfiable and the classification digest is

23298cb8c6abf989052a249db8ce48c9d73b233ef1c5b07d73054d4a130913d4.

At order 26 the source has 1,297 caps and 859 911 deletion poles. Both implementations answer all 8 599 110 boundary queries satisfiably, with classification digest

abc46bc0af6f94596ae11c68af95acc55cf8d73d82f81aea4a06ab8614a14da2.

For each order an independent verifier checks every graph premise, reconstructs every independent-edge deletion, and compares the two complete tables. Canonical source completeness again relies on Snarkhunter 2.0b.

7.6 Two explicit boundary witnesses at order 28

Theorem 7.8 (Order-28 two-orbit classification). *Every independent-edge deletion from every graph in the retained cyclically 4-edge-connected non-Tait simple cubic order-28 source realizes both fixed-five boundary words*

$$(01, 01, 01, 01) \quad \text{and} \quad (01, 01, 23, 23), \tag{7.1}$$

up to coordinate permutation. Consequently no such deletion pole has one of the six ordered exceptional exact fixed-five signatures.

Proof. In the frozen canonical ordering of the ten boundary orbits, the two words in (7.1) are orbits 0 and 2. The six exceptional masks—the three terminal-pairing orientations of each family in (6.10)—have the following literal membership table:

exact mask	orbit 0	orbit 2	excluded by
0x02b	✓	—	orbit 2
0x053	✓	—	orbit 2
0x119	✓	—	orbit 2
0x2e4	—	✓	orbit 0
0x3a4	—	✓	orbit 0
0x3c4	—	✓	orbit 0

(7.2)

An exact signature cannot equal a mask which omits a realized orbit. Thus the two positive witnesses exclude all six masks. This step uses no UNSAT inference. $\square$

The retained source was produced by

snarkhunter 28 4 S s C4 o g

and contains 12 517 graphs. This is complete for the needed cyclically four-connected cap class subject to Snarkhunter 2.0b and its documented option semantics. To see why the girth-four source loses nothing, a triangle in a simple cubic order-28 graph has three leaving edges, while its 25-vertex complement has 36 internal edges and hence contains a cycle. It would therefore expose a cyclic three-edge cut.

Every cubic order-28 source graph has 42 edges. The number of independent edge pairs in each is

$$\binom{42}{2} - 28 \binom{3}{2} = 861 - 84 = 777,$$

so the complete deletion corpus has

$$12\,517 \cdot 777 = 9\,725\,709$$

rows. A CaDiCaL-backed producer emitted a complete D_5 -labelling for each of the two words in (7.1) on every row: 19 451 418 explicit labellings and zero missing witnesses. Each certificate assigns a weight-two vector to every proper edge and semiedge. Its verification is simply the xor-zero equation at every cubic vertex plus the displayed boundary word.

The standard-library verifier reconstructs every deletion from the retained cap, checks each source for order, cubicity, connectedness, bridgelessness, triangle-freeness and non-Taitness, and verifies every displayed labelling. Its independently written zlib/C++ companion instead treats the pole and witness streams as primary input and reconstructs their incidence semantics. Both full replays pass on all 9 725 709 rows. The uncompressed pole and witness stream SHA-256 values are respectively

f42168e2786bdee409304e9e7b167ae0a0e250c78e7c2413e55ee56f5627a839,
658374b00418fa626da704670689f4dfad25cf740121509a725da4158491ef7d.

Neither verifier independently regenerates the canonical source or rechecks cyclic four-connectivity; those provenance claims rest on the retained Snarkhunter run.

Theorem 7.9 (Scoped exceptional-pole lower bound). *There is no bridge-free connected simple terminal-distinct cubic four-pole on at most 28 internal vertices whose exact fixed-five signature is one of the six ordered exceptional signatures. Hence every such pole has even order at least 30.*

In the cyclic-three branch of a reduced exceptional simple cap, the cap has order at least 30. For the equality-only and disjointness-only relations separately, it has order at least 54.

Proof. The degree sum of a terminal-distinct cubic four-pole of order n is $3n - 4$, so n is even. The complete finite cap censuses through order 22 exclude every pole in the stated scope there. Suppose an exception of order 24, 26, or 28 exists. The exact two-cut descent replaces a zero-terminal shore by one edge and a one-terminal shore by one cubic vertex; a two-plus-two composition contains a strictly smaller factor in the same exceptional family. Iterating gives, without increasing order, a vertex-minimal two-cut-reduced exceptional atom. The simple-cap reduction gives it a simple bridgeless non-Tait cubic cap $G = P + \{e, f\}$.

If G is cyclically four-edge-connected, Theorem 7.7 excludes atom orders 24 and 26, while Theorem 7.8 excludes order 28. Otherwise the structural fork proved above makes its cubic three-sum factor tree a path

$$F_1, \dots, F_k, \quad k \geq 2,$$

with e and f in the endpoint factors. Every principal cut separates the two cap edges. The rooted signatures of either shore are nonempty, and restriction to its root-end factor shows the

corresponding endpoint signature is nonempty. The root survives the deleted virtual vertex and is a nonbridge. Thus an endpoint factor of order at most 26 contains a base pair by Theorem 7.4; Theorem 6.17 lifts it to a fork on the whole shore.

If both endpoint factors had order at most 26, applying this at the two sides of any principal cut would put a fork in both shore signatures. Lemma 6.18 forces both I and D , contradicting each exceptional cross-relation

$$\{\mathsf{E}\}, \quad \{\mathsf{E}, \mathsf{I}\}, \quad \{\mathsf{D}\}.$$

Hence one endpoint factor has order at least 28. Every cubic factor has order at least four, and reversing the $k - 1$ vertex three-sums gives

$$\begin{aligned} |V(G)| &= \sum_{i=1}^k |V(F_i)| - 2(k - 1) \\ &\geq 28 + 4 + 4(k - 2) - 2(k - 1) = 2k + 26 \geq 30. \end{aligned}$$

This contradicts the assumed cap order at most 28.

For relation $\{\mathsf{E}\}$ or $\{\mathsf{D}\}$, the one-sided part of Lemma 6.18 applies at either end separately. Both endpoint factors therefore have order at least 28, and

$$|V(G)| \geq 28 + 28 + 4(k - 2) - 2(k - 1) = 2k + 50 \geq 54.$$

The lower bound 30 now follows from parity. $\square$

Order 30 remains open and may occur in either cap branch; no stronger finite lower bound is claimed. The theorem does not cover bridge-bearing poles, repeated terminals, nonsimple cores, arbitrary-colour signatures, or orientability.

8 The order-22 computational census

8.1 The exact property

For a Tait-colourable cubic graph H , define a conflict graph $\Omega(H)$ whose vertices are the edge objects of H . Join two edge objects when they are adjacent in H , or when some Tait colouring places them on the same bichromatic factor cycle.

Lemma 8.1 (Conflict-graph equivalence). *A four-edge set S is a universally separated matching if and only if it is an independent set of size four in $\Omega(H)$.*

Proof. Adjacency conflicts are exactly the negation of the matching condition. Every other conflict records one colouring and one bichromatic cycle containing two selected edges. Absence of all such conflicts is exactly separation in every Tait colouring. $\square$

Theorem 8.2 (Order-22 census). *No Tait-colourable connected simple cubic graph on 22 vertices has a universally separated four-edge matching.*

8.2 Canonical generation

The program `geng` from `nauty` and `Traces 2.9.3 [4]` was run in four modular shards:

```

geng -cq -d3 -D3 22 0/4
geng -cq -d3 -D3 22 1/4
geng -cq -d3 -D3 22 2/4
geng -cq -d3 -D3 22 3/4

```

Here `-c` requires connectedness, `-d3 -D3` requires degree three, and `geng` canonically generates simple graphs. The four disjoint shards contain respectively

$$1\,376\,411, \quad 2\,078\,782, \quad 1\,716\,645, \quad 2\,147\,609$$

graphs, for a total of 7 319 447.

8.3 Two enumeration methods

The primary program fixes the three colours at one vertex, thereby quotienting the six global colour permutations. It recursively enumerates every remaining Tait colouring, records each bichromatic factor cycle, builds $\Omega(H)$, and performs exact backtracking for an independent four-set.

The replay program does not use the primary colouring recursion. It enumerates every perfect matching M , retains M exactly when its complement is a union of even cycles, and chooses independently one of the two alternating colourings on every complementary cycle. It then records the three bichromatic factor partitions and builds the same semantic conflict graph. Each Tait colouring modulo global colour permutation is visited six times: three choices of distinguished colour matching and two orders for the other two colours.

Both methods skip a graph only after establishing that it has no Tait colouring. On every shard they agree on the graph count, the number of Tait-colourable graphs, the number of normalized Tait colourings, and the zero witness count:

Shard	Graphs	Tait-colourable	Normalized colourings	Witnesses
0	1,376,411	1,338,944	17,481,016	0
1	2,078,782	2,029,518	28,040,733	0
2	1,716,645	1,686,315	22,708,101	0
3	2,147,609	2,119,958	27,130,262	0
Total	7,319,447	7,174,735	95,360,112	0

The replay additionally visits 312 583 931 perfect matchings. By Lemma 8.1, the exhaustive zero-witness output on all 7 174 735 eligible graphs proves Theorem 8.2, conditional on the correctness of the generator and implementations.

8.4 Controls and limitations

The two implementations agree on complete lower-order corpora:

Order	Graphs	Tait-colourable	Colourings	Perfect matchings	Witnesses
12	85	80	307	902	0
16	4,060	3,848	23,420	73,929	0
18	41,301	39,687	307,606	995,256	0

These three controls were rerun from the present source during the manuscript audit.

A positive target-three control at order 24 has graph6 encoding

W'??A???A_@_A_c0?S_Gc'_?@0@@?@00??_????_??H_A?E

Both programs enumerate 36 normalized Tait colourings, return the same first separated triple (edge identifiers 0, 1, 34), and reject a separated four-set. This exercises the positive witness path, although it is not a positive four-mark control.

The computation does not address multigraphs, orders 24 and above, or the rooted avoiding-cycle property. The two programs share the mathematical conflict-graph specification and the graph6 input format; their enumeration engines are different, but they are not formal proof-producing verifiers. Reproduction of the full census is the available check of the finite claim.

9 Reproducibility

The source distribution containing this manuscript stores the two programs at

```
scratch/tait_all_coloring_mark_separation.cpp
scratch/verify_order22_separation_via_matchings.cpp
```

and the machine-readable run record at

```
scratch/order22-universal-four-separation-result.json
```

Paths are relative to the project root, one directory above this preprint directory.

The finite truth-table checks supporting the rooted algebra, though not needed by its displayed proofs, are stored at

```
scratch/rooted_cycle_translation_blockers.py
scratch/verify_rooted_cycle_translation_blockers.mjs
scratch/rooted-cycle-translation-blockers-result.json
scratch/rooted_three_pole_signature_factorization.py
scratch/verify_rooted_three_pole_signature_factorization.mjs
scratch/rooted-three-pole-signature-factorization-result.json
```

The first replay checks all $2^{10} - 1$ nonempty K_5 edge supports. The second enumerates all nonempty pairs of stabilizer-orbit unions and confirms the four equality-only, 26 ordered disjoint-only, and thirteen unordered disjoint-only cases.

The rooted order-17 package, the endpoint-factor package through order 26, and the completed cap packages are stored at

```
search/rooted-three-pole-frontier-20260727/
  order17-base-pair-report.json
  artifacts/order17-base-pair/
  verify_order17_base_pair_run.py
search/rooted-three-pole-nontait-endpoint-frontier-20260727/
  report.json
  artifacts/
  verify.py
search/four-pole-order22-cap-20260727/
  report.json
  artifacts/order22-all-cap-poles.g6.gz
  artifacts/full-run/
```

```

    verify_completed_run.py
search/four-pole-order24-cyclic4-cap-20260727/
search/four-pole-order26-cyclic4-cap-20260727/
search/four-pole-order28-cyclic4-cap-20260727/
    report.json
    artifacts/cyclic4-nontait-order28.g6
    artifacts/cyclic4-nontait-order28-poles.g6.gz
    artifacts/cadical-two-orbit-witness.tsv.gz
    verify.py
    independent-fast-replay.json
scratch/verify_boundary_two_orbit_witness_fast.cpp

```

The complete order-22 all-cap phase is included: its two classifiers agree on all eight shard digests and find zero exceptional rows among 5 956 104 deletion poles. The order-28 package retains two positive certificates per deletion row. The Python replay reconstructs all rows from the cap source; the separately written C++ replay reads the two compressed certificate streams directly. Both check every one of the 19 451 418 labellings.

The fork-crossing replay and its retained result are

```

scratch/verify_rooted_end_factor_fork_lift.py
scratch/rooted-end-factor-fork-lift-replay.json

```

with SHA-256 values

```

    verifier: 2474c8869be64e3386b23a58e1b929e5e0f56227d19033e3d4316fbb17e1f136,
    result: 738cbf30b1361b6de5bc18a13e4dfbafa9878a8bfdd9755e21b98afcda6dd513.

```

The semantic build and run commands are:

```

clang++ -O3 -std=c++20 \
    scratch/tait_all_coloring_mark_separation.cpp \
    -o /tmp/tait_all_coloring_mark_separation

geng -cq -d3 -D3 22 SHARD/4 |
    /tmp/tait_all_coloring_mark_separation \
    --target 4 --stop-after-first --progress 250000

clang++ -O3 -std=c++20 \
    scratch/verify_order22_separation_via_matchings.cpp \
    -o /tmp/verify_order22_separation_via_matchings

geng -cq -d3 -D3 22 SHARD/4 |
    /tmp/verify_order22_separation_via_matchings \
    --target 4 --stop-after-first --progress 250000

```

The frozen SHA-256 identifiers are:

Artifact	SHA-256
Primary source	a00c2f53a3b95a06a7e8767aae2eb2bfa794fc95eeeba18bcac34aea6a12072a
Replay source	104dca4ac143581a9b306172354428a96913f3fe1701c60b2a018510990eceed
geng 2.9.3 binary	ad2f68adf733dbed7cad543841cfa329740596ee5cd136f17a0a17f6e744f5ad
Primary run binary	a63d8048528e3dfa751049fbad6aec3b4909d1f7508f9e56859edbb5af9a6aa0
Replay run binary	3de46d3c343fb3b23679334f1058104488077386f49b4a2541273aa1e5d7d703
Corrected census report	53faa21acfaf75e6bf1d54d6c7d73150a57207067be4f44e20c55be220c322a2
Corrected run JSON	59d8e99cfff573de6ffa8a4df122099db3fd856d8e87803920e9cf69d6946ca8

The recorded environment was macOS 26.5.2 on arm64 with Apple clang 21.0.0. Mach-O linker UUIDs can make recompilations byte-distinct, so the source hashes and semantic counts are the portable anchors; the binary hashes identify the executables used in the archived runs. The JSON record includes all shard counts, commands, controls, and scope warnings. Its arithmetic totals were independently recomputed during this manuscript audit.

10 Relationship among the results and open boundary

Theorem 4.1 is unrooted: its binary cycle may contain a prescribed edge f . Theorem 5.1 is rooted but assumes the marks are separated in one fixed common-colour Tait colouring and leaves a cyclic four-cut interface. Neither theorem implies the other. Proposition 2.1 supplies the exact algebraic language for the later rooted interface. Theorem 6.4 then restricts a different, exceptional three-pole interface, but only conditionally on a singleton root signature; Theorem 6.10 turns a wider failure into an ordinary cut certificate. Neither is an elimination. Theorem 6.12 proves base-pair closure for the unbounded class of rooted poles whose one-vertex cap is Tait-colourable. Corollary 6.13 deliberately distinguishes the valid one-sided exclusions from the false claim that one Tait shore removes the mixed $\{E, I\}$ relation. Theorems 7.3 and 7.4 establish exact finite rooted frontiers, while Theorems 7.5 and 7.7 establish full-signature cap frontiers and Theorem 7.8 supplies the two positive orbits needed at order 28. They are not universal signature theorems. The human endpoint-fork lift and fork-crossing lemma combine these finite inputs to prove Theorem 7.9; that theorem is still only a finite scoped exceptional-pole result. Theorem 8.2 says only that a particularly strong four-mark hypothesis cannot occur at order 22 in a Tait-colourable simple cubic graph. It is diagnostic evidence, not a substitute for the missing four-cut argument.

The marked-cycle human-proof frontier left by this paper is:

Analyze the rooted four-pole obtained when the forbidden edge belongs to an independent cyclic four-edge cut, while preserving the inherited mark-parity and fixed-colouring constraints.

The distinct fixed-five signature frontier is to prove universal base-pair closure or reduce the coordinate-cut certificates beyond the completed factor order 26, and to treat cap order 30 in both the cyclic-three and cyclically four-connected branches. Nothing here classifies the published arbitrary-colour relation. The endpoint-lift proof avoids the earlier pending triangle-screen route: it fixes one labelling of the path remainder and transports all three endpoint labels by one inverse coordinate normalization. The order-28 positive witnesses then raise the scoped lower bound to 30. Even a universal resolution of the exceptional-signature problem would still leave the remaining global reductions in a Five-Cycle Double Cover proof. No claim about the orientable conjecture is made.

Acknowledgements

The author thanks the authors of the cited structural and prescribed-cycle theorems. The substantial OpenAI Codex contribution is described in the disclosure on the first page and is repeated here to avoid ambiguity: the present text and the underlying project are AI-assisted research artifacts awaiting independent human review.

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